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Methods and systems for enhancing detection of fraudulent authentication data

Abstract

A method for enhancing detection of fraudulent authentication data includes receiving, by an electronic device, data during an authentication transaction, computing a feature vector from the received data, and normalizing the feature vector. The method also includes encoding the normalized feature vector into qubits, expanding, using at least one quantum algorithm, the normalized feature vector into a high-dimensional space, and detecting in the high-dimensional space anomalies indicative of fraud based on the qubits. Furthermore, the method includes calculating, based on the detected anomalies, a confidence score reflecting a likelihood that the received data is genuine and comparing the confidence score against a threshold value. In response to determining the confidence score fails to satisfy the threshold value, the method determines that the received data requires secondary authentication.

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Background/Summary

BACKGROUND OF THE INVENTION

- (1) This invention relates generally to authentication data, and more particularly, to methods and systems for enhancing detection of fraudulent authentication data.
- (2) Known identity proofing and fraud detection techniques rely heavily on rule-based engines, neural networks, and statistical algorithms, to verify the authenticity of biometric or document data. However, the emergence of generative technologies like deep fake tools and synthetic data generators have exposed vulnerabilities in these techniques.
- (3) These techniques have been known to miss cases of synthetic or tampered identities characterized by subtle, generative adversarial network-based anomalies or morphing attempts. Additionally, these known techniques have been known to fail to capture nuanced patterns in high-dimensional biometric data, resulting in both false negatives and an inability to pinpoint sophisticated deepfake artifacts. Additionally, attackers have been increasingly using fraudulent multi-modal authentication data, for example, image data, audio data, and document image data, to increase the difficulty of detection using known techniques. Moreover, environmental noise, legitimate user variations, and constrained feature sets can cause known techniques to generate inaccurate results which undermine users' experiences and confidence.
- (4) In view of the above, it can be seen that known techniques of identity proofing and fraud detection are losing their efficacy as trustworthy and accurate methods of authenticating identities and enhancing security against attackers.
- (5) Thus, it would be advantageous and an improvement over the relevant technology to provide a method, an electronic device, and a computer-readable recording medium capable of detecting subtle, generative adversarial network-based anomalies or morphing attempts and nuanced patterns in high-dimensional biometric data, efficiently and accurately processing multi-modal authentication data, environmental noise, legitimate user variations, and constrained feature sets.

BRIEF DESCRIPTION OF THE INVENTION

- (6) In one aspect of the present disclosure, a method for enhancing detection of fraudulent authentication data including receiving, by an electronic device, data during an authentication transaction, computing a feature vector from the received data, normalizing the feature vector, and encoding the normalized feature vector into qubits. Moreover, the method includes expanding, using at least one quantum algorithm, the normalized feature vector into a high-dimensional space, detecting in the high-dimensional space anomalies indicative of fraud based on the qubits, and calculating, based on the detected anomalies, a confidence score reflecting a likelihood that the received data is genuine. The confidence score is compared against a threshold value. In response to determining the confidence score fails to satisfy the threshold value, the method determines that the received data requires secondary authentication.
- (7) In one embodiment of the present disclosure, the encoding step includes encoding the normalized feature vector using at least one of amplitude encoding and angle encoding.

- (8) In another embodiment of the present disclosure, the received data includes at least one of biometric modality data, identity document image data, and document image data.
- (9) In yet another embodiment of the present disclosure, the expanding step includes mapping features from the normalized feature vector into a high-dimensional Hilbert space.
- (10) In yet another embodiment of the present disclosure, the anomalies include synthetic biometric patterns generated by at least one of generative adversarial networks, morphing techniques, and deepfake technologies.
- (11) In yet another embodiment of the present disclosure, wherein noise-induced inaccuracies are imparted to the high-dimensional space by Noisy Intermediate-Scale Quantum hardware included in the electronic device during the expanding and detecting steps. The method further includes reducing the noise-induced inaccuracies using at least one error mitigation technique, wherein the at least one error mitigation technique comprises zero-noise extrapolation, readout error mitigation, and randomized compiling.
- (12) In yet another embodiment of the present disclosure, wherein the method further includes encrypting the received data using a post-quantum cryptographic algorithm.
- (13) Another aspect of the present disclosure provides a non-transitory computer-readable recording medium in an electronic device for enhancing detection of fraudulent authentication data. The non-transitory computer-readable recording medium stores instructions which when executed by a hardware processor, performs the steps of the methods described above.
- (14) In another aspect of the present disclosure, an electronic device for enhancing detection of fraudulent authentication data is provided that includes a processor and a memory configured to store data. The electronic device is associated with a network and the memory is in communication with the processor and has instructions stored thereon. The instructions which, when read and executed by the processor, cause the electronic device to receive data during an authentication transaction, compute a feature vector from the received data, normalize the feature vector, and encode the normalized feature vector into qubits.
- (15) Moreover, the instructions which, when read and executed by the processor, cause the electronic device to expand, using at least one quantum algorithm run by the electronic device, the normalized feature vector into a high-dimensional space, detect in the high-dimensional space, anomalies indicative of fraud based on the qubits, and calculate, based on the detected anomalies, a confidence score reflecting a likelihood that the received data is genuine. Furthermore, the instructions which, when read and executed by the processor, cause the electronic device to compare the confidence score against a threshold value, and in response to determining the confidence score fails to satisfy the threshold value, determine the received data requires secondary authentication.
- (16) In another embodiment of the present disclosure, wherein the instructions when read and executed by the processor, cause the electronic device to encode the normalized feature vector using at least one of amplitude encoding and angle encoding.
- (17) In yet another embodiment of the present disclosure, wherein the received data includes at least one of biometric modality data, identity document image data, and document image data.
- (18) In yet another embodiment of the present disclosure, wherein the instructions when read and executed by the processor, cause the electronic device to expand the feature vectors by mapping features from the normalized feature vector into a high-dimensional Hilbert space.
- (19) In yet another embodiment of the present disclosure, the anomalies include synthetic biometric patterns generated by at least one of generative adversarial networks, morphing techniques, and deepfake technologies.
- (20) In yet another embodiment of the present disclosure, wherein noise-induced inaccuracies are imparted to the high-dimensional space by Noisy Intermediate-Scale Quantum hardware included in the electronic device while expanding the normalized feature vector and detecting anomalies, and the instructions when read and executed by the processor, cause the electronic device to reduce

the noise-induced inaccuracies using at least one error mitigation technique. Wherein the at least one error mitigation technique comprises zero-noise extrapolation, readout error mitigation, and randomized compiling.

(21) In yet another embodiment of the present disclosure, the instructions when read and executed by the processor, cause the electronic device to encrypt the received data using a post-quantum cryptographic algorithm.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a side view of a person operating an example electronic device to capture authentication data;

(2) FIG. 2 is a schematic diagram of an example computing system for enhancing detection of fraudulent authentication data according to an embodiment of the present disclosure;

(3) FIG. 3 is a more detailed schematic diagram illustrating the example electronic device used for enhancing detection of fraudulent authentication data according to an embodiment of the present disclosure;

(4) FIG. 4 is a detailed schematic diagram illustrating an example quantum computer and an example quantum control system included in the computing system shown in FIG. 2, used for enhancing detection of fraudulent authentication data according to an embodiment of the present disclosure;

(5) FIG. 5 is a flowchart illustrating an example method and algorithm for encoding feature vectors using amplitude encoding according to an embodiment of the present disclosure;

(6) FIG. 6 is a flowchart illustrating an example method and algorithm for encoding feature vectors according to an embodiment of the present disclosure; and

(7) FIG. 7 is a flowchart illustrating an example method and algorithm for enhancing detection of fraudulent authentication data according to an embodiment of the present disclosure.

DETAILED DESCRIPTION OF THE INVENTION

(8) The following detailed description is made with reference to the accompanying drawings and is provided to assist in a comprehensive understanding of various example embodiments of the present disclosure. The following description includes various details to assist in that understanding, but these are to be regarded merely as examples and not for the purpose of limiting the present disclosure as defined by the appended claims and their equivalents. The words and phrases used in the following description are merely used to enable a clear and consistent understanding of the present disclosure. In addition, descriptions of well-known structures, functions, and configurations may have been omitted for clarity and conciseness. Those of ordinary skill in the art will recognize that various changes and modifications of the example embodiments described herein can be made without departing from the spirit and scope of the present disclosure.

(9) FIG. 1 is a side view of a person **10** operating an example electronic device **12** to capture authentication data, for example, during an authentication transaction. Authentication data can be any data that may be used, for example, to verify the identity of a person or verify the genuineness of, for example, an identity document. Authentication data that may be used to verify the identity of a person includes, but is not limited to, biometric modality data of the person and biometric templates created from the biometric modality data. Authentication data that may be used to verify the genuineness of an identity document includes an image of the identity document, for example, the driver's license or passport of the person.

(10) FIG. 2 is a schematic diagram of an example computing system **100** for enhancing detection of fraudulent authentication data according to an embodiment of the present disclosure. As shown in FIG. 2, the main elements of the system **100** include the electronic device **12** and a quantum

computer **14** communicatively connected via a network **16**.

(11) In FIG. 2, the electronic device **12** can be any wireless hand-held consumer electronic device capable of at least downloading applications over the Internet, running applications, capturing and storing data temporarily and/or permanently, and otherwise performing any and all functions described herein by any non-quantum computer, non-quantum computer system, non-quantum server or non-quantum electronic device included in the system **100**. Moreover, the electronic device **12** may alternatively be any type of non-quantum server or non-quantum computer implemented as a network server or network computer. Other examples of the electronic device **12** include, but are not limited to, a cellular phone, a tablet computer, a phablet computer, a laptop computer, a camera and any type of hand-held consumer electronic device having wired or wireless networking capabilities capable of performing the non-quantum functions, methods, and/or algorithms described herein.

(12) Although the electronic device **12** is described herein as performing non-quantum functions, it is contemplated by the present disclosure that the electronic device **12** may include, for example, a quantum processing unit and quantum memory such that the electronic device **12** may perform quantum functions as well as non-quantum functions.

(13) The quantum computer **14** performs quantum functions only. The quantum computer **14** and the electronic device **12** communicate via the network **16**. Such communication facilitates combining the capabilities of quantum computing with non-quantum computing to enhance the detection of anomalies in authentication data that may be indicative of synthetic fraud, deep fakes, and tampered authentication data. As a result, the accuracy and trustworthiness of authentication transaction results are facilitated to be enhanced. It is contemplated by the present disclosure that the quantum computer **14** may alternatively be remotely located in a cloud data center.

(14) The network **16** may be implemented as a 5G communications network. Alternatively, the network **16** may be implemented as any wireless network including, but not limited to, 4G, 3G, Wi-Fi, Global System for Mobile (GSM), Enhanced Data for GSM Evolution (EDGE), and any combination of a LAN, a wide area network (WAN) and the Internet. The network **16** may also be any type of wired network or a combination of wired and wireless networks.

(15) It is contemplated by the present disclosure that the number of electronic devices **12** and quantum computers **14** is not limited to the number shown in the system **100**. Rather, any number of electronic devices **12** and quantum computers **14** may be included in the system **100**.

(16) FIG. 3 is a more detailed schematic diagram illustrating the example electronic device **12** used for enhancing detection of fraudulent authentication data, for example, during an authentication transaction according to an embodiment of the present disclosure. The electronic device **12** includes components such as, but not limited to, one or more processors **18**, a memory **20**, a gyroscope **22**, an accelerometer **24**, a bus **26**, a camera **28**, a user interface **30**, a display **32**, a sensing device **34**, and a communications interface **36**. General communication between the components in the electronic device **12** is provided via the bus **26**.

(17) The processor **18** executes software instructions, or computer programs, stored in the memory **20**. It is contemplated by the present disclosure that the number of processors **18** is not limited to the number shown in the electronic device **12**. Rather, any number and type of processor(s) **18** may be included in the electronic device **12**. As used herein, the term processor is not limited to just those integrated circuits referred to in the art as a processor, but broadly refers to a computer, a microcontroller, a microcomputer, a programmable logic controller, an application specific integrated circuit, a Tensor Processing Unit (TPU), a Graphics Processing Unit (GPU), and any other programmable circuit capable of executing at least a portion of the functions and/or methods described herein. The above examples are not intended to limit in any way the definition and/or meaning of the term “processor.” The processor **18** may additionally include a quantum processing unit for performing quantum functions.

(18) The memory **20** may be any non-transitory computer-readable recording medium. Non-

transitory computer-readable recording media may be any tangible computer-based device implemented in any method or technology for short-term and long-term storage of information or data. Moreover, the non-transitory computer-readable recording media may be implemented using any appropriate combination of alterable, volatile or non-volatile memory or non-alterable, or fixed, memory. The alterable memory, whether volatile or non-volatile, can be implemented using any one or more of static or dynamic RAM (Random Access Memory), a floppy disc and disc drive, a writeable or re-writeable optical disc and disc drive, a hard drive, flash memory or the like. Similarly, the non-alterable or fixed memory can be implemented using any one or more of ROM (Read-Only Memory), PROM (Programmable Read-Only Memory), EPROM (Erasable Programmable Read-Only Memory), EEPROM (Electrically Erasable Programmable Read-Only Memory), and disc drive or the like. Furthermore, the non-transitory computer-readable recording media may be implemented as smart cards, SIMs, any type of physical and/or virtual storage, or any other digital source such as a network or the Internet from which computer programs, applications or executable instructions can be read.

(19) The memory **20** may be used to store any type of data **38**, for example, probability distributions, confidence scores, risk ratings, classifications, feature vectors, and data records of people. Each different data record is typically for a different person. The data record for each person may include data such as, but not limited to, authentication data associated with the person, biometric templates, high-dimensionality feature vectors, record high-dimensionality feature vector distributions, facial landmark data, geometric relationships between facial landmarks, and personal data.

(20) A biometric template can be any type of mathematical representation of biometric modality data. Biometric modality data is the data of a biometric modality of a person. For the methods and systems described herein, the biometric modality is face. However, it is contemplated by the present disclosure that the biometric modality may alternatively be any biometric modality that facilitates detecting fraudulent biometric modality data as described herein. Examples of other biometric modalities include, but are not limited to, iris, fingerprint, voice, palm, and behavioral patterns. Moreover, the biometric modality may be any combination of these and/or other biometric modalities including, but not limited to, the combination of voice and face and the combination of face and palm. Behavioral patterns include, but are not limited to, typing patterns and the walking gait of a person.

(21) Biometric modality data may be captured in any manner. For example, for face biometric data, the camera **28** may record image data of the face of a person by taking one or more photographs or digital images of the person, or by taking a video of the person. The camera **28** may record a sequence of digital images at irregular or regular intervals. A video is an example of a sequence of digital images being captured at a regular interval. For voice biometric data, the electronic device **12** may record a person speaking.

(22) Captured biometric modality data may be temporarily or permanently stored in the electronic device **12** or in any device capable of communicating with the electronic device **12** via the network **16**. Alternatively, the captured biometric modality data may not be stored. As used herein, capture means to record temporarily or permanently, any data including, for example, biometric modality data of a person.

(23) The term “personal data” as used herein includes any demographic information regarding a person as well as contact information pertinent to the person. Such demographic information includes, but is not limited to, the person's name, age, date of birth, street address, email address, citizenship, marital status, and contact information. Contact information can include devices and methods for contacting the person.

(24) Additionally, the memory **20** can be used to store any type of software **40**. As used herein, the term “software” is intended to encompass an executable computer program that exists permanently or temporarily on any non-transitory computer-readable recordable medium that causes, for

example, the electronic device **12** to perform at least a portion of the functions, methods, and/or algorithms described herein. Application programs are software and include, but are not limited to, operating systems, Internet browser applications, authentication applications, user liveness detection applications, an artifact detection application, a face tracker application, a feature extraction application, wavelet transforms, Principal Component Analysis (PCA) algorithms, and any other software and/or any type of instructions associated with algorithms, processes, or operations for controlling the general functions and operations of the electronic device **12**. The software may also include computer programs that implement buffers and use RAM to store temporary data.

(25) Authentication applications enable the electronic device **12** to conduct user verification and identification (1:C) transactions with any type of authentication data, where “C” is a number of candidates.

(26) Liveness detection applications differentiate between captured data of a biometric modality of a live person and manipulated biometric modality data of a person. Examples of liveness detection include, but are not limited to, analyzing captured data of a biometric modality for eye blink frequency, for replay, for subtle skin texture variations, for depth, for facial structure, and for a pulse. Using liveness detection applications facilitate ensuring that genuine biometric modality data of a person is used, for example, during authentication transactions and while remotely applying for an identity document thus enhancing the accuracy and trustworthiness of authentication transaction results as well as the trustworthiness of, for example, government issued identity documents.

(27) A machine learning algorithm (MLA) may be used to train a machine learning model (MLM) for enhancing detection of morphed biometric modality data. MLMs have parameters which are modified during training to optimize functionality of the models trained using a machine learning algorithm (MLA). The MLM may be retrained using morphed biometric modality data captured, for example, from people applying remotely for identity documents. MLAs include at least classifiers and regressors. Example classifiers are Deep Neural Networks (DNNs), Time Delay Neural Networks (TDNNs), Recurrent Neural Networks (RNNs), Convolutional Neural Networks (CNNs), Residual Networks (ResNets), Generative Adversarial Networks (GANs), transformers, and ensemble learning models.

(28) Principal Component Analysis (PCA) algorithms and wavelet transforms function to convert captured or received biometric data into a condensed feature vector. Amplitude encoding algorithms and angle encoding algorithms are encoding schemes that each function to encode feature vectors into qubits. It is contemplated by the present disclosure that the encoding scheme used for encoding a feature vector may be determined based on hardware restraints, data dimensionality, and a desired classification where the classification refers to whether the data is identified as genuine or fraudulent.

(29) The process of verifying the identity of a person is known as a verification transaction or an authentication transaction. Generally, during a verification transaction a biometric template is generated from biometric modality data of a person captured during the transaction. Typically, data for a single biometric modality is captured. The generated biometric template is compared against a corresponding record biometric template of the person and a matching score is calculated for the comparison. If the matching score meets or exceeds a threshold score, the identity of the person is verified as true. Alternatively, the captured biometric modality data may be compared against corresponding record biometric modality data to verify the identity of the person

(30) Known methods of biometric authentication have difficulty detecting subtle modifications made to genuine authentication data. The subtle modifications are imperceptible to humans. Providing such modified authentication data while applying, for example, for an identity document is known as a morphing attack.

(31) Convolutional Neural Networks, for example, may be used to extract high-dimensionality feature vectors from biometric modality data captured, for example, during an authentication

transaction or while a person applies remotely for an identity document. High dimensionality feature vectors are numerical representations of, for example, biometric modality data that include hundreds or thousands of dimensions. Each dimension corresponds to a distinct feature of the biometric modality. Using high dimensionality feature vectors facilitates the detailed analysis necessary for detecting anomalies that facilitate distinguishing between genuine and morphed biometric modality data.

(32) Convolutional Neural Networks may also be used, for example, to extract high-dimensionality feature vectors from an identity document image captured, for example, while a person remotely establishes a banking account.

(33) High-dimensionality feature vectors may be created from different types of biometric modality data. For example, for facial image data, high dimensionality feature vectors typically range between about 128 and 512 dimensions but have been known to include up to 1,024 dimensions to capture finer details of the face. For iris data, high dimensionality feature vectors typically range between about 200 and 400 dimensions which represent intricate iris patterns. For voice data, high dimensionality feature vectors typically range between about 50 and 600 dimensions which facilitate analyzing the variability in, for example, pitch, tone, and resonance of the voice data. High dimensionality feature vectors created from multi-modal biometric data typically include thousands of dimensions which are necessary to obtain distinct details for each biometric modality.

(34) A Gaussian Mixture Model (GMM), for example, may be used to generate the distribution of the high-dimensionality feature vectors for captured biometric modality data. Any significant deviation of the generated distribution from a record distribution may indicate that the biometric modality data includes anomalies indicative of morphing. A significant deviation between the distribution of captured high-dimensionality feature vectors and the record distribution can indicate that the biometric data likely contains anomalies consistent with morphing.

(35) Morphed data often distorts geometric relationships. As a result, any anomaly or difference detected between the calculated and record geometric relationships may indicate the facial image data was morphed.

(36) When a sequence of digital images is captured, the electronic device **12** may extract images from the sequence and assign a time stamp to each extracted image. An application, for example a face tracker application may process the extracted digital images. The face tracker application may extract data from images including, but not limited to, facial landmarks, facial expressions, and the orientation of the face within the image. The extracted data may be analyzed to determine whether the image or images include anomalies indicative of morphing. For example, the optical flow of facial landmarks may be tracked across sequential video frames to identify unnatural transitions that may be indicative of morphing. Such transitions may be anomalies indicative of morphing.

(37) Facial action patterns of the landmarks may also be analyzed to facilitate detecting anomalies in image data. Facial action patterns refer to dynamic changes in facial expressions and movements, analyzed through Action Units (AUs) such as eyebrow raises, lip pursing, and cheek lifts. Unnatural transitions or inconsistencies in these patterns, detected via optical flow analysis or machine learning, may be anomalies indicative of morphing. Additionally, a Recurrent Neural Network (RNN), for example, may be used to model temporal dependencies to facilitate detecting anomalies indicative of morphing such as, but not limited to, motion inconsistencies in micro-expressions or facial movements over time.

(38) The facial landmark data includes data generated by the face tracker application as a result of processing the frames. The generated data includes, but is not limited to, coordinate values for facial landmarks. The coordinate values may include a three-dimensional coordinate value for each different facial landmark. The facial landmarks are identified by the face tracker application on the facial image included in each processed frame. The facial landmarks include, but are not limited to, cheek points, nose points, points on sides of the face, chin points, and points about the eyes and eyebrows.

(39) The facial landmark data may also be used to facilitate detecting anomalies in image data. For example, facial image landmark data may be used to monitor the position of each facial landmark between frames. Additionally, the facial landmark data may be used to calculate geometric relationships between facial landmarks in the same image, for example, the geometric relationships between the tip of the nose and a point on the chin or the distance between the center of the eyes. The calculated geometric relationships can be compared against corresponding record geometric relationships for the person from whom the facial image data was captured. Morphed data often distorts geometric relationships. As a result, any anomaly or difference detected between the calculated and record geometric relationships may indicate the facial image data was morphed.

(40) A temporal differential analysis of the facial landmark data may be conducted to facilitate determining whether any landmarks moved too rapidly between frames which may indicate the sequence of digital images was morphed. Too rapidly can mean moving to a different position within a subsequent frame, but the movement is physically impossible. Moreover, a three-dimensional analysis of each facial landmark may be conducted to facilitate determining whether or not the sequence of digital images was morphed.

(41) When the sequence of digital images includes an audio signal, feature extraction computer programs can process the audio signal to generate a representative feature vector that contains information about the signal. Audio signals can be voice data spoken by a person. Features may be extracted from voice data and analyzed using a trained machine learning model to determine whether the voice data includes anomalies indicative of morphing.

(42) Synthetic speech may mimic the intonation, rhythm, and emotional nuances of genuine human speech. As a result, synthetic speech may be a threat to security systems that use voice data to authenticate or verify the identities of people.

(43) Anomalies may be present in each characteristic that may be associated with synthetic speech. For example, for the range of pitch an anomaly may be that the received voice biometric data has a narrower range of pitch than typically included in authentic speech. For timbre, anomalies can include, but are not limited to, a lack of expected complexity, unusual harmonic structures, and erratic formant movements. For intensity or loudness, an anomaly may be variations in volume that do not correspond with an expressed or expected emotion. For voice resonators, an anomaly may be less variability and responsiveness. For pace, an anomaly may be unnatural timing patterns like a consistent speech rate or abnormal pauses. For prosody, anomalies can include inconsistencies in stress patterns or intonation curves unusual for the context or language norm. For rhythm, anomalies can include unusual pauses or changes in the rhythm of speech such as hesitations or rushed sequences. For natural speech, anomalies can include, but are not limited to, a lack of natural pitch variation across sentences, an unexpected pitch contour within a phrase, unusually long or short durations, or a lack of variability in durations. For frequency, an anomaly can be that the frequency does not exceed a threshold established for synthetic speech, for example, up to 5,354 Hz.

(44) Wavelet transforms can be used to decompose a digital image into multiple frequency bands which facilitate analyzing high and low frequency artifacts in the image. For example, a digital image may be processed by a wavelet transform to decompose the image into high, low and medium frequency artifacts. High-frequency artifacts capture fine details, such as edges and textures, while low-frequency artifacts represent broader, smoother areas of the image. The artifacts imparted to digital images via manipulation associated with morphing are typically high-frequency artifacts. Examples of high-frequency artifacts associated with morphing include, but are not limited to, texture inconsistencies, edge sharpness variations, and pixel level anomalies caused by compression or resampling during morphing. Examples of texture inconsistencies include, but are not limited to, blending irregularities in skin patterns or hairlines. Examples of edge sharpness variations include, but are not limited to, unnatural transitions between facial features.

(45) Wavelet transforms facilitate isolating these high-frequency artifacts to identify unnatural

patterns that would not be in an unaltered image. For example, the high-frequency artifacts of a received image may be compared against record high-frequency artifacts of a corresponding bona fide image. Any differences may be the result of morphing. Thus, fine details of the image are checked for manipulation that can be indicative of morphing. Wavelet transforms may generate statistical anomalies, for example, unnatural spectral patterns. Such anomalies may be indicative of morphing.

(46) The gyroscope **26** and the one or more accelerometers **24** generate data regarding rotation and translation of the electronic device **12** that may be communicated to the processor **18** and the memory **20** via the bus **26**. The gyroscope **22** and accelerometer **24** are typically included in electronic devices **12** that are primarily mobile, for example, smart phones and other smart devices, but not in electronic devices **12** that are primarily stationary, for example, servers or personal computers. Thus, the electronic device **12** may alternatively not include the gyroscope **22** or the one or more accelerometers **24** or may not include either.

(47) The camera **28** captures image data. The camera **28** can be one or more imaging devices configured to record image data of at least a portion of the body of a user including any biometric modality of the user while utilizing the electronic device **12**. The camera **28** may also capture digital images of printed images.

(48) The camera **28** is capable of recording image data under any lighting conditions including infrared light. The camera **28** may be integrated into the electronic device **12** as one or more front-facing cameras and/or one or more rear facing cameras that each incorporates a sensor, for example and without limitation, a CCD or CMOS sensor. Alternatively, the camera **28** can be external to the electronic device **12**. As used herein, capture means to record temporarily or permanently, any data including, for example, biometric modality data of a person.

(49) The user interface **30** and the display **32** allow interaction between a user and the electronic device **12**. The display **32** may include a visual display screen or monitor that displays information. For example, the display **32** may be a Liquid Crystal Display (LCD), an active-matrix display, plasma display, or cathode ray tube (CRT). The user interface **30** may include a keypad, a keyboard, a mouse, an illuminator, a signal emitter, a microphone, and/or speakers.

(50) Moreover, the user interface **30** and the display **32** may be integrated into a touch screen display. Accordingly, the display **32** may also be used to show a graphical user interface, which can display various data and provide “forms” that include fields that allow for the entry of information by the user. Touching the screen at locations corresponding to the display of a graphical user interface allows the person to interact with the electronic device **12** to enter data, change settings, control functions, etc. Consequently, when the touch screen is touched, the user interface **30** communicates this change to the processor **18**, and settings can be changed, or user entered information can be captured and stored in the memory **20**. The display **32** may function as an illumination source to apply illumination to an object while image data for the object is captured.

(51) The sensing device **34** may include Radio Frequency Identification (RFID) components or systems for receiving information from other devices in the system **100** and for transmitting information to other devices in the system **100**. The sensing device **34** may alternatively, or additionally, include components with Bluetooth, Near Field Communication (NFC), infrared, or other similar capabilities. Communications between the electronic device **12** of the user and the quantum computer **14** may occur via NFC, RFID, Bluetooth or the like only so a network connection from the electronic device **12** is unnecessary.

(52) The communications interface **36** may include various network cards, and circuitry implemented in software and/or hardware to enable wired and/or wireless communications with other electronic devices **12** (not shown) and the quantum computer **14** via the network **16**. Communications include, for example, conducting cellular telephone calls and accessing the Internet over the network **16**. By way of example, the communications interface **36** may be a digital subscriber line (DSL) card or modem, an integrated services digital network (ISDN) card, a cable

modem, or a telephone modem to provide a data communication connection to a corresponding type of telephone line. As another example, the communications interface **36** may be a local area network (LAN) card (e.g., for Ethernet™ or an Asynchronous Transfer Model (ATM) network) to provide a data communication connection to a compatible LAN. As yet another example, the communications interface **36** may be a wire or a cable connecting the electronic device **12** with a LAN, or with accessories such as, but not limited to, other electronic devices. Further, the communications interface **36** may include peripheral interface devices, such as a Universal Serial Bus (USB) interface, a PCMCIA (Personal Computer Memory Card International Association) interface, and the like.

(53) The communications interface **36** also allows the exchange of information across the network **16**. The exchange of information may involve the transmission of radio frequency (RF) signals through an antenna (not shown). Moreover, the exchange of information may be between the electronic device **12** and the quantum computer **14**, other electronic devices (not shown), and other computer systems (not shown) capable of communicating over the network **16**.

(54) Examples of other computer systems (not shown) include computer systems of service providers such as, but not limited to, financial institutions, medical facilities, national security agencies, merchants, and authenticators. The electronic devices (not shown) may be associated with any user or with any type of entity including, but not limited to, commercial and non-commercial entities.

(55) FIG. **4** is a detailed schematic diagram illustrating the example quantum computer **14** and an example quantum control system **14-1** used for enhancing detection of fraudulent authentication data, for example, during an authentication transaction according to an embodiment of the present disclosure. The quantum computer **14** and quantum control system **14-1** are typically proximate each other. It is contemplated by the present disclosure that the quantum computer **14** and quantum control system **14-1** may alternatively be remotely located in a cloud data center.

(56) The quantum computer **14** includes components such as a quantum processing unit **42**, a quantum memory **44**, noisy intermediate scale quantum (NISQ) **46**, a communications interface **48**, a user interface **50**, and a display **52**. The quantum control system **14-1** typically includes components such as, but not limited to, analog-to-digital converters, digital-to-analog converters, and field programmable gate arrays. The components of the quantum computer **14** typically do not communicate using a bus as in non-quantum computers.

(57) Transmissions from, for example, the electronic device **12** to the quantum computer **14** are received by the quantum control system **14-1**. The quantum control system **14-1** translates the transmissions into real physical signals for the qubits. The quantum processing unit **42** transmits results, for example, detected anomalies to the quantum control system **14-1**, which transmits the results to, for example, the electronic device **12**.

(58) The quantum processing unit **42** manipulates qubits instead of bits. Qubits as used herein refers to quantum states. The quantum processing unit **42** receives quantum gate instructions from the quantum control system **14-1** and uses qubits to perform computations.

(59) The quantum memory **44** may store any type of data **54** similar to the data **32** that may be stored in the memory **20** of the electronic device **12**. The quantum memory **44** may also store software **56** for performing quantum functions. Such software includes, but is not limited to, amplitude encoding algorithms, angle encoding algorithms, Quantum Support Vector Machines (QSVM), Variational Quantum Circuits (VQC), error mitigation software, and Post Quantum Cryptography (PQC) Algorithms.

(60) The NISQ **46** is hardware that can be characterized by limited qubit counts, higher error rates, and non-trivial decoherence. The NISQ **46** can run error mitigation software.

(61) Amplitude encoding algorithms enable encoding a feature vector created, for example, by the electronic device **12** into qubits or a quantum state. Angle encoding algorithms also enable encoding a feature vector created, for example, by the electronic device **12** into qubits or a quantum

state.

(62) Quantum Support Vector Machines (QSVM) facilitate detecting anomalies in authentication data. Feature vectors include floating point numbers that can represent features extracted from authentication data, for example, biometric modality data. The floating point numbers may be referred to herein as features. More specifically, QSVMs map features in feature vectors into a high-dimensional Hilbert space using quantum kernels, which facilitates enhancing the separation of genuine versus fraudulent authentication data. VQCs employ parameterized quantum circuits that facilitate detecting anomalies in the high-dimensional Hilbert space indicative of deep fakes or morphed authentication data. Using QSVM and VQC software enables enhanced anomaly detection compared against contemporary techniques using non-quantum hardware and software.

(63) In quantum mechanics, all states of a quantum system can be represented as vectors in a Hilbert space. Hilbert spaces are complex, high-dimensional vector spaces. Unlike non-quantum feature spaces, where each dimension corresponds to a single variable, a quantum state can exist as a superposition of basis states. Authentication data can be encoded into qubits using quantum computing. Each additional qubit effectively doubles the size of the state space, allowing for exponentially greater representational capacity compared to that available using non-quantum feature spaces. As a result, QSVMs and VQCs can reveal subtle correlations and anomalies by analyzing the enlarged state space. As a result, in the context of, for example, authentication transactions, genuine biometric data can be more effectively separated from fraudulent or morphed biometric data to thus enhance the detection of anomalies that contemporary techniques using non-quantum hardware and software would likely miss.

(64) Error mitigation software includes, but is not limited to, zero-noise extrapolation software, read out error mitigation software, and randomized compiling which may be used during QSVM and VQC operations to facilitate reducing noise-induced inaccuracies in anomaly detection.

(65) PQC algorithms facilitate encrypting authentication data during transmission, for example, from the quantum computer **14** to the electronic device **12** via the network **16**. Moreover, PQC algorithms facilitate encrypting stored biometric data and safeguarding sensitive data against quantum capable adversaries to ensure long term data integrity. PQC algorithms include, but are not limited to, lattice-based encryption algorithms. Encryption provided by PQC algorithms may extend to feature vectors transmitted from the electronic device **12** to the quantum computer **14** via the network **16** or may be limited to final outputs of the quantum computer **14** and stored biometric records.

(66) It is contemplated by the present disclosure that the quantum functions described herein may be performed by a quantum computer, for example the quantum computer **14**. Although the electronic device **12** is described herein as performing non-quantum functions, it is contemplated by the present disclosure that the electronic device **12** may include, for example, a quantum processing unit and quantum memory such that the electronic device **12** may perform quantum functions as well as non-quantum functions.

(67) It is contemplated by the present disclosure that the electronic device **12**, quantum computer **14** and any other computer devices (not shown) and/or systems (not shown) that may be in the computing system **100** may be implemented in a cloud environment.

(68) The communications interface **48**, user interface **50**, and display **52** are similar to the communications interface **36**, interface **30**, and display **32**, respectively, described herein with regard to the electronic device **12**.

(69) FIG. 5 is a flowchart illustrating an example method for encoding feature vectors using amplitude encoding according to an embodiment of the present disclosure. The electronic device **12** can implement the non-quantum software instructions while a person operating the electronic device **12** captures biometric modality data from his or herself during, for example, an authentication transaction. FIG. 5 illustrates example operations performed when the electronic device **12** runs non-quantum software **40** stored in the memory **20** and the quantum computer **14**

receives quantum gate instructions from the quantum control system **14-1** and uses qubits to perform computations.

(70) In step **S1**, the software **40** executed by the processor **18** causes the electronic device **12** to capture biometric modality data of the person during, for example, an authentication transaction and to create a feature vector from the captured biometric modality data. The feature vector may be expressed as $X=(x_{\text{sub.1}}, x_{\text{sub.2}} \dots x_{\text{sub.n}})$, where $x_{\text{sub.1}}, x_{\text{sub.2}} \dots x_{\text{sub.n}}$ are floating point numbers that can represent features extracted from the captured biometric modality data. Next, in step **S2**, the software **40** executed by the processor **18** causes the electronic device **12** to normalize the feature vector according to the equation $\sum x_{\text{sub.i}}^2=1$ and to transmit the normalized feature vector to the quantum computer **14** via the network **16**.

(71) In step **S3**, the quantum computer **14** creates qubits from the floating point numbers such that each floating point number becomes the amplitude of a corresponding qubit. Thus, each floating point number represents an amplitude in the quantum state, ensuring that $\sum x_{\text{sub.i}}^2=1$. For example, the corresponding qubit for the floating point $x_{\text{sub.i}}$ could be $|i\rangle$ custom character. As another example, a two-dimensional vector $X=(x_{\text{sub.1}}, x_{\text{sub.2}})$ can be mapped to as $x_{\text{sub.1}}|0\rangle$ custom character + $x_{\text{sub.2}}|1\rangle$ custom character. The thus created qubits represent an encoded version of the feature vector X .

(72) FIG. **6** is a flowchart illustrating an example method for encoding feature vectors using angle encoding according to an embodiment of the present disclosure. The electronic device **12** can implement the non-quantum software instructions while a person operating the electronic device **12** captures biometric modality data from his or herself during, for example, an authentication transaction. FIG. **6** illustrates example operations performed when the electronic device **12** runs non-quantum software **40** stored in the memory **20** and the quantum computer **14** receives quantum gate instructions from the quantum control system **14-1** and uses qubits to perform computations.

(73) In step **S4**, the software **40** executed by the processor **18** causes the electronic device **12** to capture biometric modality data of the person during, for example, an authentication transaction and to create a feature vector from the captured biometric modality data. The feature vector may be expressed as $X=(x_{\text{sub.1}}, x_{\text{sub.2}} \dots x_{\text{sub.n}})$, where $x_{\text{sub.1}}, x_{\text{sub.2}} \dots x_{\text{sub.n}}$ are floating point numbers that can represent features extracted from the captured biometric modality data. Next, in step **S5**, the software **40** executed by the processor **18** causes the electronic device **12** to normalize the feature vector according to the equation $\sum x_{\text{sub.i}}^2=1$ and to transmit the normalized feature vector to the quantum computer **14** via the network **16**.

(74) In step **S6**, the quantum computer **14** creates qubits from the floating point numbers such that each floating point number becomes a rotation angle of a corresponding qubit. Thus, each floating point number represents a rotation angle $\theta_{\text{sub.i}}$.

(75) Typically, a qubit is rotated about an axis, for example, the Y or Z-axis. Next, in step **S7**, the quantum computer **14** applies the rotations $R_{\text{sub.y}}(\theta_{\text{sub.i}})$ to the respective qubits. For example, applying $R_{\text{sub.y}}(\theta_{\text{sub.i}})$ to $|0\rangle$ custom character yields

(76) $\cos(-\frac{i}{2}) \cdot \text{Math. } 0 \cdot \text{Math. } + \sin(-\frac{i}{2}) \cdot \text{Math. } 1 \cdot \text{Math. } \dots$

(77) It should be understood that while amplitude encoding adjusts amplitudes of quantum states directly, angle encoding uses rotation gates to embed feature values. This distinction ensures the correct quantum state is created based on the chosen encoding scheme.

(78) Known identity proofing and fraud detection techniques rely heavily on rule-based engines, neural networks, and statistical algorithms, to verify the authenticity of biometric or document data. However, the emergence of generative technologies like deep fake tools and synthetic data generators have exposed vulnerabilities in these techniques.

(79) These techniques have been known to miss cases of synthetic or tampered identities characterized by subtle, generative adversarial network-based anomalies or morphing attempts. Additionally, these known techniques have been known to fail to capture nuanced patterns in high-dimensional biometric data, resulting in both false negatives and an inability to pinpoint

sophisticated deepfake artifacts. Additionally, attackers have been increasingly using fraudulent multi-modal authentication data, for example, image data, audio data, and document image data, to increase the difficulty of detection using known techniques. Moreover, environmental noise, legitimate user variations, and constrained feature sets can cause known techniques to generate inaccurate results which undermine users' experiences and confidence.

(80) In view of the above, it can be seen that known techniques of identity proofing and fraud detection are losing their efficacy as trustworthy and accurate methods of authenticating identities and enhancing security against attackers.

(81) To solve the above problems, the electronic device **12** may receive biometric modality data of a person during an authentication transaction, compute a feature vector from the received data, and normalize the feature vector. The electronic device **12** may transmit via the network **16** the normalized feature vector to the quantum computer **14** via which encodes the normalized feature vector into qubits. The quantum computer **14** may use at least one quantum algorithm to expand the normalized feature vector into a high-dimensional space, and detect in the high-dimensional space anomalies indicative of fraud based on the qubits. The detected anomalies may be transmitted from the quantum computer **14** via the network **16** to the electronic device **12**. The electronic device **12** may use the detected anomalies to calculate a confidence score reflecting the likelihood that the received biometric modality data is genuine and compare the confidence score against a threshold value. In response to determining the confidence score fails to satisfy the threshold value, the electronic device **12** can determine that the received biometric modality data requires secondary authentication.

(82) FIG. 7 is a flowchart illustrating an example method and algorithm for enhancing the detection of fraudulent authentication data during, for example, an authentication transaction according to an embodiment of the present disclosure. When a person desires to conduct an activity, the person may be required to prove his or her identity before being permitted to conduct the activity. Example activities include, but are not limited to, remotely conducting a financial transaction, remotely applying to open an account or enroll in a service, and entering a country as part of border security. FIG. 7 illustrates example operations performed when the electronic device **12** runs software **40** stored in the memory **20** and the quantum computer **14** receives quantum gate instructions from the quantum control system **14-1** and uses qubits to perform computations.

(83) In step **S8**, the software **40** executed by the processor **18** causes the electronic device **12** to receive authentication data associated with a person. For example, the authentication data may be captured by another electronic device (not shown) in the system **100** and transmitted via the network **16** to the electronic device **12** for receipt by the electronic device **12**. Alternatively, the person may operate the electronic device **12** to capture authentication data. As a result of capturing the authentication data the electronic device **12** also receives the authentication data. In this example method the authentication data is facial biometric data of a person. The facial biometric data may be an image of the person's face. The biometric modality may alternatively be any modality that may be used to verify the identity of the person as described herein. The received authentication data may be processed after receipt to ensure image quality. For example, the received authentication data may be processed to reduce noise, for image enhancement, and dimensionality reduction to ensure image quality.

(84) In step **S9**, the software **40** executed by the processor **18** causes the electronic device **12** to compute a feature vector from the received facial biometric data. The software **40** that may be used to compute the feature vector includes, for example, principal component analysis models and wavelet transforms. In step **S10**, the software **40** executed by the processor **18** causes the electronic device **12** to normalize the feature vector and transmit via the network **16** the normalized feature vector to the quantum computer **14**.

(85) Next, in step **S11**, the quantum computer **14** encodes the normalized feature vector into qubits, or quantum states. The normalized feature vector may be encoded into qubits using, for example,

amplitude or angle encoding techniques as described herein with regard to FIGS. 5 and 6, respectively.

(86) In step S12, the quantum computer 14 expands the normalized feature vector into a high dimensional space, for example, a Hilbert high dimensional space. Next, in step S13, the quantum computer 14 detects in the high dimensional space anomalies indicative of fraud based on the qubits and transmits via the network 16 the detected anomalies to the electronic device 12. It is contemplated by the present disclosure that the quantum computer 14 encrypts the detected anomalies before transmission to the electronic device 12. Such quantum software includes, for example, post quantum cryptography algorithms.

(87) Quantum Support Vector Machines (QSVM) facilitate detecting anomalies in authentication data. Feature vectors include floating point numbers that can represent features extracted from authentication data, for example, biometric modality data. The floating point numbers may be referred to herein as features. More specifically, QSVMs map features in feature vectors into a high-dimensional Hilbert space using quantum kernels, which facilitates enhancing the separation of genuine versus fraudulent authentication data. VQCs employ parameterized quantum circuits that facilitate detecting anomalies in the high-dimensional Hilbert space indicative of deep fakes or morphed authentication data. Using QSVM and VQC software enables enhanced anomaly detection compared against contemporary techniques using non-quantum hardware and software.

(88) In quantum mechanics, all states of a quantum system can be represented as vectors in a Hilbert space. Hilbert spaces are complex, high-dimensional vector spaces. Unlike non-quantum feature spaces, where each dimension corresponds to a single variable, a quantum state can exist as a superposition of basis states. Authentication data can be encoded into qubits using quantum computing. Each additional qubit effectively doubles the size of the state space, allowing for exponentially greater representational capacity compared to that available using non-quantum feature spaces. As a result, QSVMs and VQCs can reveal subtle correlations and anomalies by analyzing the enlarged state space. As a result, in the context of, for example, authentication transactions, genuine biometric data can be more effectively separated from fraudulent or morphed biometric data to thus enhance the detection of anomalies that contemporary techniques using non-quantum hardware and software would likely miss.

(89) In step 14, the software 40 executed by the processor 18 causes the electronic device 12 to calculate, based on the detected anomalies, a confidence score reflecting the likelihood that the received data is genuine. Next, in step S15, the software 40 executed by the processor 18 causes the electronic device 12 to compare the confidence score against a confidence threshold value. If the confidence score satisfies the confidence threshold value, in step S16, the software 40 executed by the processor 18 causes the electronic device 12 to determine that the received facial biometric data is genuine. However, when the confidence score fails to satisfy the confidence threshold value, in step S17, the software 40 executed by the processor 18 causes the electronic device 12 to determine that the received facial biometric data may not be genuine and as a result that secondary authentication is required. Secondary authentication includes, but is not limited to, manual review or other contemporary biometric authentication methods.

(90) The confidence threshold value described herein may be satisfied when the confidence score is greater than or equal to the confidence threshold value. Other confidence threshold values may be satisfied when the calculated confidence score is less than or equal to the confidence threshold value. Alternatively, the confidence threshold value may include multiple threshold values, each of which is required to be satisfied to satisfy the confidence threshold value.

(91) Although the example method and algorithm for enhancing the detection of fraudulent data describes transmitting detected anomalies to the electronic device 12, the quantum computer 14 may additionally calculate and transmit a performance score. The performance score can reflect, for example, that the detected anomalies correspond to the received biometric modality and the number of detected anomalies. If an adequate number of anomalies is not detected and/or the anomalies do

not correspond to the received biometric modality, accurate and trustworthy authentication transaction results cannot be generated. The adequate number of detected anomalies may be determined, for example, based on expected patterns for the modality of received biometric data or using a statistically based anomaly number threshold value. Should the number of detected anomalies satisfy the anomaly number threshold value, the number of detected anomalies can be adequate; otherwise, not.

(92) The anomaly number threshold value described herein may be satisfied when an anomaly score based on the number of anomalies is greater than or equal to the anomaly number threshold value. Other anomaly number threshold values may be satisfied when the anomaly score is less than or equal to the anomaly number threshold value. Alternatively, the anomaly number threshold value may include multiple anomaly number threshold values, each of which is required to be satisfied to satisfy the anomaly number threshold value.

(93) The performance score can be generated, for example, by comparing the number, nature, or distribution of detected anomalies against an expected baseline for the particular biometric modality. For example, the quantum computer **14** might assess how closely the detected anomalies match known morphing or synthetic patterns typical of that modality. A higher degree of mismatch can lower the performance score, indicating insufficient or irrelevant anomaly detection. Conversely, if the anomalies align well with what the modality analysis anticipates, the performance score increases. This calculation ensures that not only the quantity of anomalies is considered, but also their consistency with the received data type. Tracking the performance score enables deciding whether more anomalies are needed or if a different approach to detecting anomalies should be used.

(94) Thus, the performance score can be used to improve the performance of the quantum computer **14**. For example, the detected anomalies and calculated performance score can be transmitted to the electronic device **12**, which determines whether the performance of the computer **14** is satisfactory based on the performance score. The electronic device **12** may compare the performance score against a performance threshold value. If the performance score satisfies the performance threshold value, the electronic device **12** calculates the confidence score in step **S15**. Otherwise, the electronic device **12** can transmit the normalized feature vectors and feedback to the quantum computer **14**.

(95) Feedback may include, for example, that more anomalies are required, that the anomalies do not correspond to the received biometric modality, or both. In response, the quantum computer **14** again detects anomalies and calculates the performance score and transmits the detected anomalies and performance score to the electronic device **12**. It is contemplated by the present disclosure that the electronic device **12** and the quantum computer **14** may repeatedly communicate as described above until a satisfactory performance score is calculated. Such repeated communications between the electronic device **12** and the quantum computer **14** reflect the synergy between them.

(96) The performance threshold value described herein may be satisfied when the performance score is greater than or equal to the performance threshold value. Other performance threshold values may be satisfied when the performance score is less than or equal to the performance threshold value. Alternatively, the performance threshold value may include multiple performance threshold values, each of which is required to be satisfied to satisfy the performance threshold value.

(97) It is contemplated by the present disclosure that the synergy between non-quantum and quantum computing enables sub-second or near-real-time anomaly detection. Although current NISQ hardware may introduce overhead for data encoding and qubit measurement, the combination of non-quantum and quantum computing described herein facilitates maintaining low-latency performance by leveraging fast non-quantum pre-processing and error-mitigation techniques that minimize the duration of quantum operations. For example, in high-throughput environments such as border checkpoints or financial transactions, the method described herein

with regard to FIG. 7 may be executed in under a few seconds, effectively blocking fraudulent attempts without noticeable delay. As quantum gate fidelities and qubit counts continue to improve, latencies will further decrease, causing instantaneous or near-instantaneous accurate and trustworthy authentication transaction results.

(98) Using the methods and algorithms for enhancing detection of fraudulent authentication data described herein facilitates seamlessly combining the capabilities of quantum computing and non-quantum computing to enhance the detection of deep fakes, synthetic speech and morphing attacks. As a result, limitations of purely non-quantum computing techniques are facilitated to be overcome and the accuracy and trustworthiness of authentication transactions and fraud detection results are facilitated to be enhanced.

(99) It is contemplated by the present disclosure that the example methods and algorithms described herein may be conducted entirely by the electronic device **12**; partly by the electronic device **12** and partly by the quantum computer **14**; entirely by the quantum computer **14**, or by any other combination of other servers (not shown), electronic devices (not shown), or computers (not shown) operable to communicate with the electronic device **12** and the quantum computer **14** via the network **16**. Furthermore, data described herein as being stored in the electronic device **12** may alternatively, or additionally, be stored in any other server (not shown), electronic device (not shown), or computer (not shown) operable to communicate with the electronic device **12** via the network **16**.

(100) Additionally, the example methods and algorithms described herein may be implemented with any number and organization of computer program components. Thus, the methods and algorithms described herein are not limited to specific computer-executable instructions. Alternative example methods and algorithms may include different computer-executable instructions or components having more or less functionality than described herein.

(101) The example methods and/or algorithms described above should not be considered to imply a fixed order for performing the method and/or algorithm steps. Rather, the method and/or algorithm steps may be performed in any order that is practicable, including simultaneous performance of at least some steps. Moreover, the method and/or algorithm steps may be performed in real time or in near real time. It should be understood that for any method and/or algorithm described herein, there can be additional, fewer, or alternative steps performed in similar or alternative orders, or in parallel, within the scope of the various embodiments, unless otherwise stated. Furthermore, the invention is not limited to the embodiments of the methods and/or algorithms described above in detail.

Claims

1. A method for enhancing detection of fraudulent authentication data comprising the steps of: receiving, by a non-quantum electronic device, data during an authentication transaction; computing a feature vector from the received data; normalizing the feature vector; transmitting, via a network, the normalized feature vector to a quantum computer; encoding, by the quantum computer, the normalized feature vector into qubits; expanding, by the quantum computer, using at least one quantum algorithm, the qubits into a high-dimensional space; detecting, by the quantum computer, in the high-dimensional space anomalies indicative of fraud based on the qubits; transmitting, by the quantum computer, via the network, the detected anomalies to the non-quantum electronic device; calculating, by the non-quantum electronic device based on the detected anomalies, a confidence score reflecting a likelihood that the received data is genuine; comparing the confidence score against a threshold value; and in response to determining the confidence score fails to satisfy the threshold value, determining the received data requires secondary authentication.
2. The method according to claim 1, said encoding step comprising encoding the normalized feature vector using at least one of amplitude encoding and angle encoding.

3. The method according to claim 1, wherein the received data comprises at least one of: biometric modality data; identity document image data; and document image data.
4. The method according to claim 1, said expanding step comprising applying the at least one quantum algorithm to the qubits to map the normalized feature vector into a high-dimensional Hilbert space.
5. The method according to claim 1, wherein the anomalies comprise synthetic biometric patterns generated by at least one of generative adversarial networks, morphing techniques, and deepfake technologies.
6. The method according to claim 1, wherein noise-induced inaccuracies are imparted to the high-dimensional space by Noisy Intermediate-Scale Quantum hardware included in the quantum computer during said expanding and detecting steps, said method further comprising reducing the noise-induced inaccuracies using at least one error mitigation technique, wherein the at least one error mitigation technique comprises zero-noise extrapolation, readout error mitigation, and randomized compiling.
7. The method according to claim 1, further comprising encrypting the received data using a post-quantum cryptographic algorithm.
8. A non-quantum electronic device for enhancing detection of fraudulent authentication data comprising: a processor; and a memory configured to store data, said non-quantum electronic device being associated with a network and said memory being in communication with said processor and having instructions stored thereon which, when read and executed by said processor, cause said non-quantum electronic device to: receive data during an authentication transaction; compute a feature vector from the received data; normalize the feature vector; transmit, via the network, the normalized feature vector to a quantum computer, wherein the quantum computer encodes the normalized feature vector into qubits, expands, using at least one quantum algorithm the qubits into a high-dimensional space, detects in the high-dimensional space anomalies indicative of fraud based on the qubits, and transmits the detected anomalies via the network to said non-quantum electronic device; calculate, based on the detected anomalies, a confidence score reflecting a likelihood that the received data is genuine; compare the confidence score against a threshold value; and in response to determining the confidence score fails to satisfy the threshold value, determine the received data requires secondary authentication.
9. The non-quantum electronic device according to claim 8, wherein the normalized feature vector is encoded using at least one of amplitude encoding and angle encoding.
10. The non-quantum electronic device according to claim 8, wherein the received data comprises at least one of: biometric modality data; identity document image data; and document image data.
11. The non-quantum electronic device according to claim 8, wherein the feature vector is expanded by applying the at least one quantum algorithm to the qubits to map the normalized feature vector into a high-dimensional Hilbert space.
12. The non-quantum electronic device according to claim 8, wherein the anomalies comprise synthetic biometric patterns generated by at least one of generative adversarial networks, morphing techniques, and deepfake technologies.
13. The non-quantum electronic device according to claim 8, wherein noise-induced inaccuracies are imparted to the high-dimensional space by Noisy Intermediate-Scale Quantum hardware included in the quantum computer while expanding the normalized feature vector and detecting anomalies, wherein the noise-induced inaccuracies are reduced using at least one of zero-noise extrapolation, readout error mitigation, and randomized compiling.
14. The non quantum electronic device according to claim 8, wherein the instructions when read and executed by said processor, cause said non quantum electronic device to encrypt the received data using a post-quantum cryptographic algorithm.
15. A non-transitory computer-readable recording medium in a non-quantum electronic device for enhancing detection of fraudulent authentication data, the non-transitory computer-readable

recording medium storing instructions which when executed by a hardware processor cause the non-transitory recording medium to perform steps comprising: receiving data during an authentication transaction; computing a feature vector from the received data; normalizing the feature vector; transmitting, via a network, the normalized feature vector to a quantum computer, wherein the quantum computer encodes the normalized feature vector into qubits, expands, using at least one quantum algorithm the qubits into a high-dimensional space, detects in the high-dimensional space anomalies indicative of fraud based on the qubits, and transmits via the network the detected anomalies to said non-quantum electronic device; calculating, based on the detected anomalies, a confidence score reflecting a likelihood that the received data is genuine; comparing the confidence score against a threshold value; and in response to determining the confidence score fails to satisfy the threshold value, determining the received data requires secondary authentication.

16. The non-transitory computer-readable recording medium according to claim 15, wherein the normalized feature vector is encoded using at least one of amplitude encoding and angle encoding.

17. The non-transitory computer-readable recording medium according to claim 15, wherein the received data comprises at least one of: biometric modality data; identity document image data; and document image data.

18. The non-transitory computer-readable recording medium according to claim 15, wherein the at least one quantum algorithm is applied to the qubits to map the normalized feature vector into a high-dimensional Hilbert space to expand the normalized feature vector into the high-dimensional space.

19. The non-transitory computer-readable recording medium according to claim 15, wherein noise-induced inaccuracies are imparted to the high-dimensional space by Noisy Intermediate-Scale Quantum hardware included in the quantum computer while expanding the normalized feature vector and detecting anomalies indicative of fraud, and the noise-induced inaccuracies are reduced using at least one of zero-noise extrapolation, readout error mitigation, and randomized compiling.

20. The non-transitory computer-readable recording medium according to claim 15, wherein the instructions when read and executed by said processor, cause said non-transitory computer-readable recording medium to encrypt the received data using a post-quantum cryptographic algorithm.
